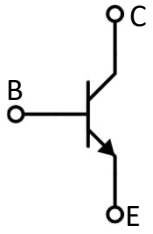
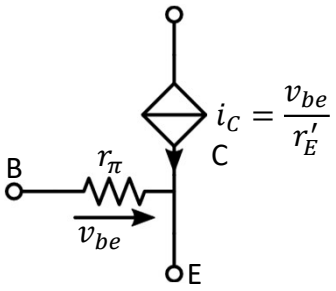
# Résumé Amplificateurs BJT
# Zusammenfassung BJT-Verstärker

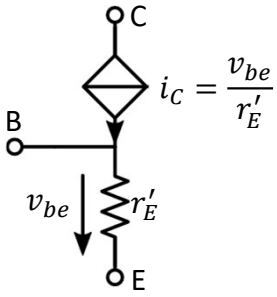
Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

*Modèle AC petits signaux / AC* Kleinsignalmodell

NPN



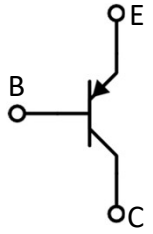



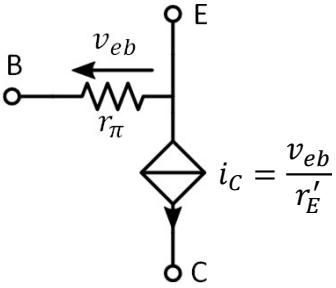



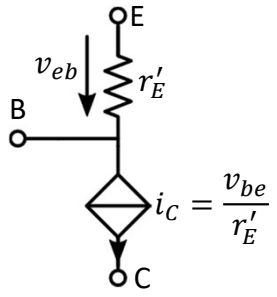

$$r'_E = \frac{V_T}{I_{C0}}$$

$$r_\pi = \beta r'_E = \frac{\beta V_T}{I_{C0}}$$

PNP

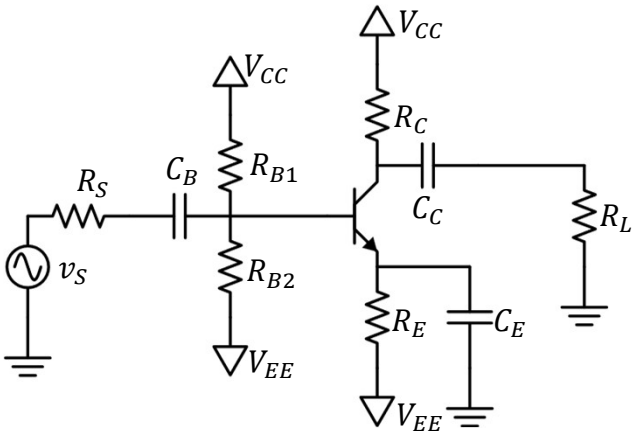


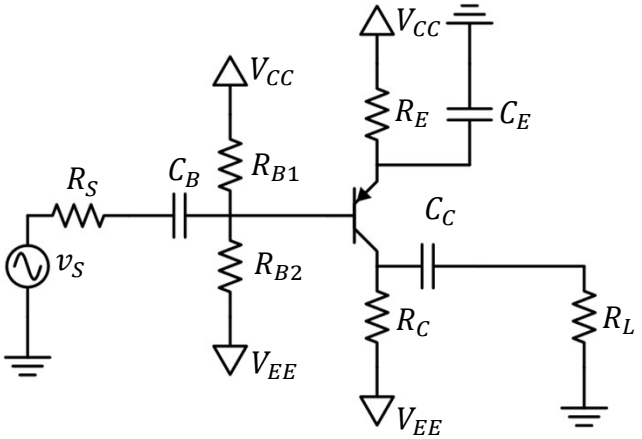



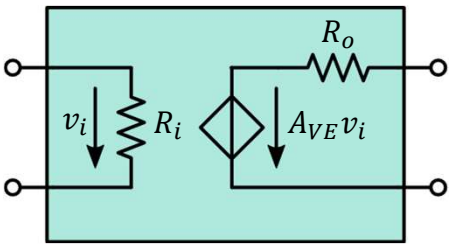




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Ampli Emetteur Commun (CE) / Verstärker in Emitterschaltung (CE)

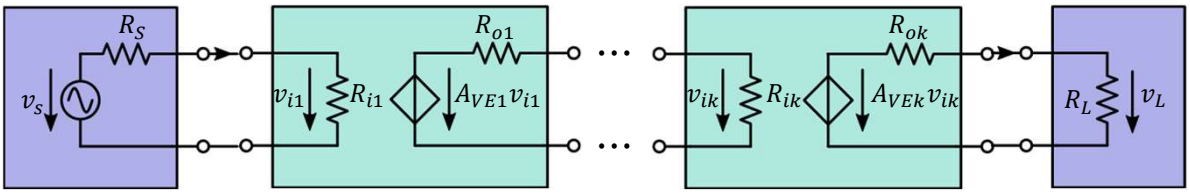



$$R_i = R_B \parallel r_\pi$$
$$A_{VE} = -\frac{R_C}{r'_E}$$
$$R_O = R_C$$

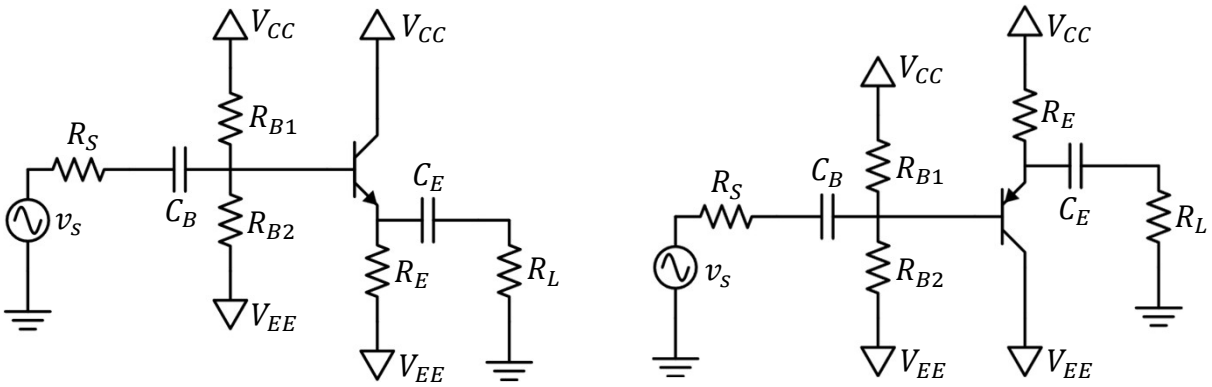
## Amplificateurs cascadés / Kaskadierte Verstärker




$$A_V = \frac{v_L}{v_s} = \frac{R_{i1}}{R_{i1}+R_S} A_{VE1} \frac{R_{i2}}{R_{o1}+R_{i2}} A_{VE2} \cdots \frac{R_{ik}}{R_{o(k-1)}+R_{ik}} A_{VEk} \frac{R_L}{R_{ok}+R_L}$$

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## Ampli Collecteur Commun (CC) / Verstärker in Kollektorschaltung (CC)




$$r_{in} = \beta(r'_E + R_E \parallel R_L) \parallel R_B \approx \beta R_E \parallel \beta R_L \parallel R_B$$

$$A_V = \frac{v_L}{v_s} = \frac{r_{in}}{r_{in}+R_S} \frac{R_E \parallel R_L}{r'_E + R_E \parallel R_L} \approx \frac{r_{in}}{r_{in}+R_S}$$